
PO12Q - Quantitative Political Analysis: Uncovering Relationships

Week 2 - Worksheet Solutions

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The Correlation Widget – Exercises

1. Increasing n from 50 to 300 makes the sample r cling more tightly to the target (≈ 0.60), narrows its confidence interval, and yields a much smaller p -value. This shows that larger samples reduce sampling variability and increase power.
2. With a small true effect ($r \approx 0.20$) and $n = 60$, the observed r can bounce widely across seeds and may even flip sign occasionally. This instability comes from high sampling error relative to the weak signal.
3. Often yes: r can be near zero despite a clear U-shape. The positive and negative slopes on either side of the minimum/maximum cancel in a linear summary, masking the strong but non-linear association.
4. When Noise is large enough that the two arms blur together (threshold depends on n and the scale), both the straight line and LOESS become close to flat and similarly uninformative.
5. Yes. The straight line averages across the curve and often understates the visible relationship (sometimes showing a small slope). LOESS follows the bend and reveals a strong monotonic trend.
6. As Noise increases, r typically declines quickly toward zero because curvature plus added noise erodes linear association. The LOESS curve still shows the underlying shape longer, revealing curvature even when r is small.
7. For a curved monotonic pattern, the straight line can imply weak or even misleading direction, while LOESS shows the true increasing pattern with curvature (steeper ends, flatter middle).
8. In Linear mode (truly linear data), LOESS and the least-squares line essentially coincide, especially as n grows, because the best smooth for a straight-line signal is a straight line.
9. With $r = 0.40$ and $n = 80$, r typically varies noticeably across seeds (often on the order of ± 0.1 or so). With $n = 400$, the variability shrinks substantially (often to just a few hundredths), illustrating how larger n stabilizes estimates.
10. For Cubic with low Noise, r is moderately stable but still more variable than in a truly linear case because the linear summary is sensitive to where the curve is sampled. Stability improves with larger n and lower Noise.

11. You can tune Cubic (noise) to get a Pearson r similar to a Linear scenario (e.g., both around 0.5). The plots differ: Linear looks elliptical with a constant slope; Cubic is curved. LOESS highlights this curvature, showing why the same r can represent different relationships.
12. Pearson's r is adequate when the relationship is approximately linear and residual variation is roughly symmetric and constant. Prefer a non-linear description when you see curvature or non-monotonic structure that a straight line cannot capture.
13. A non-linear model is justified when LOESS departs clearly from the straight line, residuals show systematic patterns, and a non-linear fit improves predictive performance while matching the visible shape in the plot.

Correlation by Hand

1.
 - a. FALSE: A coefficient of -1 indicates a perfect negative correlation. No correlation would be indicated with a correlation coefficient of zero.
 - b. TRUE: Pearson's correlation assumes a linear relationship, whereas Spearman's relies only on the relationship being monotonic – one variable consistently increasing or decreasing as the other changes, though not necessarily at a constant rate or in a straight line. Spearman's is therefore better suited to non-linear monotonic relationships.
 - c. TRUE: The variance is the covariance of a variable with itself: setting $y = x$ in the covariance formula returns the formula for the variance.
 - d. FALSE: Covariance is not bounded between -1 and $+1$. Its value depends on the measurement units and variability of the two variables, so there is no universal fixed range for covariance. It is the correlation coefficient, obtained by dividing the covariance by the product of the variables' standard deviations, that is standardised and bounded between -1 and $+1$.
 - e. TRUE: Pearson's correlation is calculated from the means and standard deviations of the raw data, both of which can be heavily affected by extreme values, making it sensitive to outliers.
 - f. TRUE: Spearman's coefficient first ranks the values of each variable and then applies Pearson's correlation to those ranks.
 - g. TRUE: The point-biserial correlation is a special case of Pearson's correlation, applied when one variable is continuous and the other is binary (coded 0/1).
 - h. FALSE: The ϕ -coefficient measures the association between two dichotomous/binary variables, not two continuous ones. For two continuous variables we would use Pearson or Spearman.
 - i. TRUE: A coefficient of $+1$ indicates perfect positive correlation, meaning all observations lie exactly on a straight (upward-sloping) line.
 - j. FALSE: The slope of the line does not matter. The correlation coefficient measures how tightly the observations cluster around a line, not how steep that line is.
 - k. FALSE: A significant correlation establishes association only. Correlation is necessary but not sufficient for causation, so it can never demonstrate that one variable causes another on its own.

2.
 - a. Spearman's correlation — the outliers in the mark make Pearson unreliable, whereas Spearman's, being rank-based, is robust to extreme values (and makes no normality assumption).
 - b. Spearman's correlation — both variables are ordinal.
 - c. Point-biserial correlation — one continuous variable (GDP) and one dichotomous variable (regime: democratic/autocratic).
 - d. Pearson's correlation — two continuous, approximately normally distributed variables.
 - e. ϕ -correlation — two dichotomous variables (school type and admission yes/no).
3. We have data on how many seminars a student has attended and their final exam score.
 - a. Intermediate calculations:

$$\bar{x} = \frac{7 + 4 + 2 + 9 + 6}{5} = 5.6, \quad \bar{y} = \frac{65 + 48 + 38 + 90 + 80}{5} = 64.2$$

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
7	65	1.4	0.8	1.12	1.96	0.64
4	48	-1.6	-16.2	25.92	2.56	262.44
2	38	-3.6	-26.2	94.32	12.96	686.44
9	90	3.4	25.8	87.72	11.56	665.64
6	80	0.4	15.8	6.32	0.16	249.64
Sum				215.40	29.20	1864.80

$$r_p = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} = \frac{215.40}{\sqrt{29.20 \times 1864.80}} = \frac{215.40}{233.35} = 0.92$$

b.

$$t = \frac{r_p \sqrt{n-2}}{\sqrt{1-r_p^2}} = \frac{0.923\sqrt{3}}{\sqrt{1-0.923^2}} = 4.16$$

With $n - 2 = 3$ degrees of freedom, the critical value for a two-tailed test at 95% confidence is 3.182. Since $4.16 > 3.182$, the coefficient is statistically significant.

- c. The coefficient is positive and very strong ($r_p = 0.92$): students who attended more seminars tended to achieve higher final exam scores.

4. Consider the data on the place achieved in a swimming competition and the number of hours spent training per week.

a. Ranks and squared rank differences:

x	y	Rank(x)	Rank(y)	d_i	d_i^2
14	1	5	1	4	16
8	2	3	2	1	1
7.5	3	2	3	-1	1
9	4	4	4	0	0
7	5	1	5	-4	16
Sum					34

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} = 1 - \frac{6 \times 34}{5(5^2 - 1)} = 1 - \frac{204}{120} = -0.70$$

b.

$$t = \frac{r_s \sqrt{n-2}}{\sqrt{1-r_s^2}} = \frac{-0.70\sqrt{3}}{\sqrt{1-(-0.70)^2}} = \frac{-0.70\sqrt{3}}{\sqrt{1-0.49}} = -1.70$$

With $n - 2 = 3$ degrees of freedom, the critical value for a two-tailed test at 95% confidence is 3.182. Since $|-1.70| < 3.182$, the coefficient is not statistically significant.

- c. The relationship is not statistically significant at the 95% confidence level, and so we have insufficient evidence to reject the null hypothesis of no relationship between training time and performance.

5. We have data on whether a country has a first-past-the-post (FPTP) electoral system (yes/no) and its average satisfaction score with democracy (percent).

a. Group means:

$$\bar{y}_1 = \frac{95 + 80}{2} = 87.5$$

$$\bar{y}_0 = \frac{80 + 75 + 82}{3} = 79$$

This gives us the two sub-sample sizes:

$$n_1 = 2$$

$$n_0 = 3$$

$$n = 5$$

The overall mean of satisfaction is:

$$\bar{y} = \frac{80 + 75 + 95 + 80 + 82}{5} = 82.4$$

Intermediate calculation for the standard deviation of the continuous variable:

y	$y - \bar{y}$	$(y - \bar{y})^2$
80	-2.4	5.76
75	-7.4	54.76
95	12.6	158.76
80	-2.4	5.76
82	-0.4	0.16
Sum		225.20

$$s_y = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n - 1}} = \sqrt{\frac{225.20}{4}} = \sqrt{56.30} = 7.50$$

Correlation coefficient:

$$r_{pb} = \frac{\bar{y}_1 - \bar{y}_0}{s_y} \times \sqrt{\frac{n_1 n_0}{n(n-1)}}$$

$$r_{pb} = \frac{87.5 - 79}{7.50} \times \sqrt{\frac{2 \times 3}{5(4)}}$$

$$r_{pb} = \frac{8.5}{7.50} \times \sqrt{\frac{6}{20}}$$

$$r_{pb} = 1.13 \times 0.55 = 0.62$$

b.

$$t = \frac{r_{pb} \sqrt{n-2}}{\sqrt{1-r_{pb}^2}}$$

$$t = \frac{0.62\sqrt{3}}{\sqrt{1-0.62^2}}$$

$$t = \frac{0.62\sqrt{3}}{\sqrt{1-0.38}}$$

$$t = 1.37$$

With $n - 2 = 3$ degrees of freedom, the critical value for a two-tailed test at 95% confidence is 3.182. Since $1.37 < 3.182$, the coefficient is not statistically significant.

c. The relationship is not statistically significant at the 95% confidence level, and so we have insufficient evidence to reject the null hypothesis of no relationship between electoral system and democratic satisfaction.

6. a. With $A = 10$, $B = 5$, $C = 8$, and $D = 12$:

$$\phi = \frac{AD - BC}{\sqrt{(A + B)(C + D)(A + C)(B + D)}}$$

$$\phi = \frac{(10)(12) - (5)(8)}{\sqrt{(10 + 5)(8 + 12)(10 + 8)(5 + 12)}}$$

$$\phi = \frac{120 - 40}{\sqrt{(15)(20)(18)(17)}}$$

$$\phi = \frac{80}{\sqrt{91,800}}$$

$$\phi = \frac{80}{302.99} = 0.26$$

- b.

$$\chi^2 = n\phi^2$$

$$\chi^2 = 35 \times 0.26^2$$

$$\chi^2 = 35 \times 0.0697 = 2.44$$

With 1 degree of freedom, the critical value for 95% confidence is 3.84. Since $2.44 < 3.84$, the association is not statistically significant.

- c. The relationship is not statistically significant at the 95% confidence level, and so we have insufficient evidence to reject the null hypothesis of no relationship between populist leaders and economic growth

7. Interpret each of the following correlation coefficients, using **cohen:2013**<empty citation>'s strength thresholds for correlation coefficients:
- a. $r_p = -0.74, p \leq 0.05$ – a large, negative effect: more cigarettes smoked per year is strongly associated with lower life expectancy (in years)
 - b. $r_{pb} = -0.31, p \leq 0.05$ – a medium, negative effect: being a gym member (yes/no) is associated with a lower number of classes attended
 - c. $r_s = 0.34, p \leq 0.05$ – a medium, positive effect: higher satisfaction with a module (1-5) is associated with a higher class of marks (Fail, Third, 2.2, 2.1, First)
 - d. $\phi = 0.1, p \leq 0.05$ – a small, positive effect: receiving free school meals (yes/no) is weakly associated with entering higher education (yes/no)